

Benchmarking Forecasting Systems Under Friction: When Forecast Rankings Misselect Deployed Winners

Abstract

Forecasting systems are commonly benchmarked by forecast-side scores, yet under deployment frictions these rankings need not preserve deployed operational ranking. We identify a benchmark failure mode: under a fixed deployed interface, the forecast-side winner can recurrently fail to be the deployed winner once friction is moderate or high and proposed actions are not directly realized. We separate two questions: a mechanism question about whether the same proposal yields different realized outcomes under different interfaces (Q1), and a benchmark-selection question about whether forecast ranking still selects the deployed best system under one fixed interface (Q2). Across a controlled synthetic anchor, a deliberately small forecasting-native event-micro benchmark, and operational inventory corroboration, zero-friction alignment is exact or near-stable, while moderate-to-high friction produces repeated ranking drift and deployed misselection. Forecast-side metrics may remain valid for prediction, yet under a fixed frictional deployment interface they need not be the right object for model selection.

1 Introduction

A forecasting benchmark is unreliable for deployment if the model that wins on forecast-side scores recurrently fails to be the model that performs best after actions are realized under friction. Forecast-side metrics can be fully appropriate for measuring predictive accuracy, yet they need not be the right object for model selection once forecasts are translated into actions through a frictional interface. This is not a claim that forecast metrics are incorrect; it is a claim that they may be the wrong object for deployment-time model selection under friction. In such settings, forecasts do not directly determine realized outcomes: they first induce proposed actions, which are then transformed by operational rules, adjustment costs, or switching penalties before realized performance is observed.

This paper studies two evaluation questions for forecasting-driven systems under friction. Q1 isolates the mechanism: if the same proposed action path is held fixed, can different interfaces produce different realized outcomes? Q2 asks the benchmark question: under one fixed deployed interface, can forecast-side ranking still be trusted to select the operationally best forecasting system? Forecast-side metrics may remain valid for prediction, yet under a fixed frictional deployment interface they need not be the right object for model selection.

This paper makes three narrow contributions. First, it identifies a benchmark failure mode for forecasting systems under a fixed deployed interface: at moderate-to-high friction, forecast-side ranking can fail to preserve deployed ranking. Second, it shows that this is not merely descriptive ranking drift: in operational settings, moderate-to-high friction can produce recurrent deployed misselection. Third, it separates mechanism (Q1) from benchmark consequence (Q2), clarifying why realized-action evaluation can diverge from forecast-side or target-side quantities and when that divergence matters for model selection. We support this claim with a controlled synthetic anchor, a deliberately small forecasting-native event-micro benchmark, operational corroboration from inventory, and exact-control corroboration from portfolio Q1 support.

2 Two Evaluation Questions

Let $\hat{y}_t$ denote a forecast, a_t^{tar} a proposed action derived from it, and a_t^{exec} the realized action after a frictional interface is applied. Let $U(a, y)$ denote realized operational utility against the realized outcome y_t . We distinguish three evaluation objects: forecast-side quality, target-side decision quality, and realized-action quality. The benchmark question is which of these objects is appropriate for model selection under deployment constraints.

Q1 (same proposal, different interface). Holding the proposal fixed at $a_{1:T}^{tar}$, we ask whether varying only the interface that maps proposals to realized actions changes realized operational outcomes. This is an exact-control question: proposal identity must be locked, so any downstream difference must arise from the interface rather than from a different forecast or proposal source.

Q2 (fixed interface, different forecasts). Q2 is the benchmark-selection question. Holding the interface fixed, we ask whether varying the forecaster changes the relationship between forecast-metric ranking and deployed operational ranking. More directly, does ranking systems by forecast quality still identify the system that is best under deployed realized-action evaluation?

These questions are complementary but not interchangeable. Q1 explains why divergence can arise once realized actions depart from proposed targets, while Q2 asks whether that divergence becomes a benchmark model-selection failure under a fixed deployed interface. Because the two questions change different parts of the pipeline, evidence for one should not be read as automatic evidence for the other.

3 Evidence Design

Our evidence hierarchy separates sanity anchoring, forecasting-native main evidence, operational corroboration, and corroborating support. Synthetic provides the zero-friction sanity anchor and controlled ranking-drift evidence. A deliberately small event-micro benchmark provides the main forecasting-native Q2 evidence: models emit event

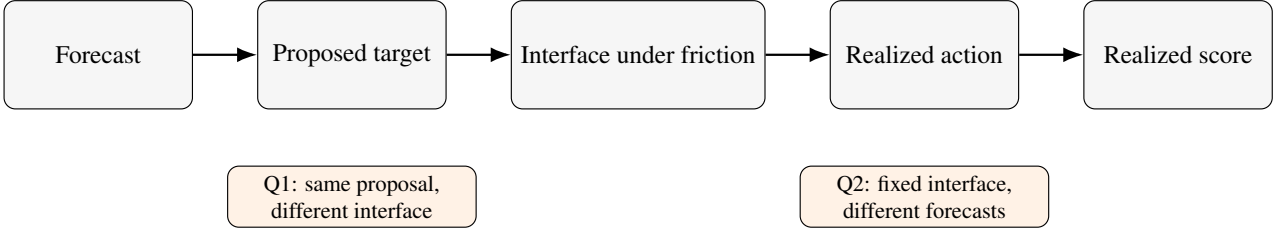


Figure 1: Forecasts are converted into proposed target actions and then translated into realized actions through an interface operating under friction, which creates two distinct evaluation questions: whether the same frozen proposal can yield different realized outcomes under different interfaces (Q1), and whether forecast-metric ranking matches deployed operational ranking under a fixed interface (Q2). This separation changes the evaluation object and therefore can change both benchmark interpretation and deployed model selection under friction.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.70	0.022	3/10
0.25	Small MLP	Linear AR	0.70	0.053	3/10
0.50	Small MLP	Moving average (7)	0.10	0.308	9/10
1.00	Small MLP	Moving average (7)	0.00	1.187	10/10

Table 1: Forecast-side vs. deployed-side model selection under a fixed replenishment interface. Winner columns report the most frequent seed-level best model at each friction level. Positive mean deployed gap indicates that the forecast-selected model underperforms the deployed-selected model. Deployed-suboptimal seeds / total counts seeds in which the forecast-selected model is not deployed-best.

probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. Inventory provides operational corroboration, where the same benchmark consequence appears with stronger recurrence magnitudes. Portfolio is used only for exact-control Q1 corroboration, where proposal identity can be locked directly. Load-following remains appendix-only operational support.

In inventory Q2, the locked main table compares five forecasting families under one responsive replenishment rule: Naive persistence, Moving average (7), Linear AR, Small MLP, and Small GRU. The appendix widens that comparison with validation-selected family representatives under the same split, held-out protocol, validation criterion, and standardized two-stage tuning procedure; simple heuristic baselines remain fixed ex ante. The main text keeps the locked five-family comparison, while the appendix records the broader family sweep for baseline-specification transparency.

4 Results

4.1 Q2: Fixed interface, benchmark failure and selection consequence

Q2 asks the paper’s headline benchmark question: under one fixed deployed interface, can forecast-side ranking still be trusted for deployed model selection? Synthetic provides the zero-friction sanity anchor: rankings align exactly at zero friction, and once friction is moderate or high, repeated pairwise flips and lower rank correlation emerge.

Figure 2 and Table 2 provide the paper’s main forecasting-native evidence block. In the event-micro benchmark, agreement is 0.75 at friction 0.00, falls to 0.30 by 0.50, and reaches 0.05 at 1.00. Over the same rows, the deployed winner shifts from Reactive sharp to Calibrated baseline and then Lagged smoother, while deployed suboptimality rises from 5/20 seeds at zero friction to 14/20 at 0.50 and 19/20 at 1.00. Appendix

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.75	0.002	5/20
0.50	Reactive sharp	Calibrated baseline	0.30	0.013	14/20
1.00	Reactive sharp	Lagged smoother	0.05	0.061	19/20

Table 2: Forecast-side vs. deployed-side model selection in the forecasting-native event-micro benchmark. Under a fixed thresholding interface with switching friction, the forecast-selected model becomes recurrently deployed-suboptimal as friction rises.

Tables A7 and A8 show that the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists under $\tau = 0.50$ and under log loss.

Table 1 remains the paper’s main operational benchmark table. Inventory provides operational corroboration with stronger recurrence magnitudes. The friction 0.25 row is best read as a transition regime: the most frequent deployed winner changes to Linear AR, yet agreement remains 0.70, deployed suboptimality remains 3/10, and the median deployed gap remains 0.0. By friction 0.5, however, benchmark failure is already recurrent: the forecast-selected model is deployed-suboptimal in 9/10 seeds, agreement falls to 0.10, and the median deployed gap reaches 0.227. At friction 1.0, deployed suboptimality rises to 10/10 seeds, with mean and median deployed gaps of 1.187 and 1.162. Thus the practical issue is not merely that rankings drift, but that model choice under a fixed deployed rule becomes repeatedly wrong at moderate-to-high friction.

Appendix Tables A3–A5 widen the inventory comparison set under a standardized tuning protocol. That broader-family sweep strengthens moderate/high-friction mismatch, but the zero- and low-friction rows remain too mixed for headline use.

4.2 Q1: Same proposal, different interface

Q1 is the mechanism section: it explains why the Q2 benchmark failure can arise. In the synthetic benchmark, holding the proposal fixed while varying only the interface yields exact zero-friction coincidence and a proposal-versus-realization gap that rises from 0.000 at zero friction to 0.223 at friction 1.0.

Inventory reproduces the same mechanism in an operational setting: zero friction is exact by construction, low friction can remain mixed, and sufficiently high friction makes tempered execution beneficial through lower adjustment costs. Portfolio provides exact-control corroboration only: proposal identity is directly verified, zero-cost behavior is near-flat, and positive-cost improvements recur across universes. Q1

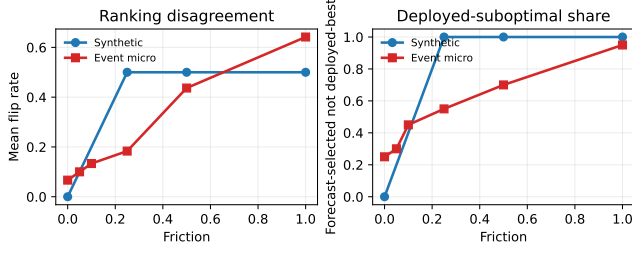


Figure 2: Q2 — synthetic provides the zero-friction sanity anchor, while event micro provides the main forecasting-native evidence line under a fixed thresholding interface. The left panel shows pairwise ranking disagreement (mean flip rate); the right panel shows the operational benchmark-consequence quantity, the share of seeds in which the forecast-selected model is not deployed-best. Table 1 then shows the same consequence in the operational inventory domain, with stronger recurrence magnitudes.

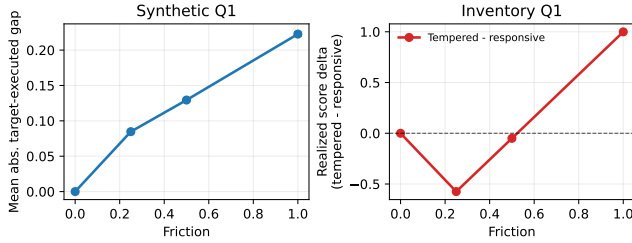


Figure 3: Q1 — the same proposal can yield different realized outcomes under different interfaces. At zero friction, target-side and realized-side quantities coincide or nearly coincide. Under positive friction, interface-mediated differences emerge. The synthetic benchmark shows the controlled phenomenon cleanly, while the inventory domain shows a threshold pattern: low friction can remain mixed, but sufficiently high friction makes the tempered interface beneficial through reduced adjustment costs. The two panels use different y-axis quantities because the synthetic and inventory Q1 evidence emphasize controlled gap emergence and operational score effects, respectively.

therefore explains why forecast-side, target-side, and realized-action evaluation can diverge, while Q2 shows when that divergence becomes a model-selection problem.

5 Discussion and Limitations

The paper’s practical claim is narrow: forecast-side metrics may remain valid for prediction, yet under a fixed frictional deployment interface they need not be the right object for model selection. In the forecasting-native event-micro benchmark, that mismatch becomes a recurrent benchmark-selection failure at moderate-to-high friction, and inventory shows the same consequence in an operational domain with stronger recurrence magnitudes. Q1 explains why divergence can arise, and Q2 shows when that divergence matters for model selection.

Three limitations bound the claim. Forecasting-native Q2 evidence comes from a deliberately small event-micro benchmark. Operational corroboration remains concentrated in one inventory domain. In the expanded-family inventory sweep,

moderate/high-friction mismatch strengthens but zero- and low-friction rows remain mixed, so we do not use that block as main-text confirmation.

The resulting implication is evaluative rather than predictive: forecasting systems under deployment constraints should be judged with realized-action-aware criteria rather than forecast-side quantities alone.

A Evidence Hierarchy

Domain	Question	Fixed	Varying	Zero-fric.	Positive-fric.	Role
Synthetic	Q1	proposal	interface	exact agreement	mismatch emerges	required
Synthetic	Q2	interface	forecaster	aligned at zero	ranking flips	sanity anchor
Inventory	Q1	proposal	interface	coincide at zero	threshold story	required
Inventory	Q2	interface	forecaster	transition at 0.25	stronger recurrence	operational corrob.
Portfolio	Q1	proposal	interface	mostly near-flat	positive-cost recurrence	support
Event micro	Q2	interface	forecaster	near-stable at zero	recurrence selection failure recurs	main forecast evidence

Table A1: Evidence map for the two evaluation questions, covering the synthetic, inventory, portfolio, and event-forecasting evidence blocks used in the paper and appendix.

B Additional Q2 Package Details

This appendix records the same-interface rank-based package after the fifth inventory baseline is added and keeps the synthetic result visible.

Portfolio is omitted from Q2 here because it does not satisfy the zero-friction sanity condition after the fifth baseline is added.

C Stronger Inventory Q2 Baselines

To reduce baseline-specification ambiguity, we widen the inventory Q2 comparison set with stronger classical and neural families while leaving the environment, split, friction grid, fixed responsive interface, deployed metric, and tie policy unchanged. All tunable families share the same held-out protocol, validation criterion, and standardized two-stage tuning structure, while the two simple heuristics remain fixed ex ante. Family representatives are selected by validation negative MAE only, and the held-out forecast-side winners in the robustness table are then computed from the same held-out forecast metric used in the locked inventory Q2 pipeline.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A5: Expanded-family inventory Q2 robustness summary for the broader comparison set.

In this broader comparison, agreement is mixed at 0.00 and 0.25, falls further by 0.50 and 1.00, deployed-suboptimal counts rise from 4/10 to 9/10, and moderate/high-friction mismatch strengthens.

D Minimal Event-Forecasting Micro-Benchmark

We additionally construct a deliberately small binary-event setting in which each forecaster emits event probabilities, a fixed

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	pass	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed gate
Inventory	required	fail	10	0.120	0.840	1.0000	0.500	Small MLP / Moving average (7)	1.000	flip > 0.10
Portfolio	stretch	excluded	3	0.611	-0.200	0.0005	0.778	EWMA (20) / Rolling mean (20)	1.000	flip > 0.10; rho < 0.50

Table A2: Additional paper-facing Q2 package summary after the fifth inventory baseline is added.

Family	Input	Stage 1	Stage 2	Seeds	Representative rule
Naive persistence	fixed heuristic	–	–	final 10	fixed ex ante
Moving average (7)	fixed heuristic	–	–	final 10	fixed ex ante
Linear AR	legacy tabular	5	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small MLP	legacy tabular	18	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small GRU	sequence (lookback 28)	6	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Regularized linear + lag search	lagged tabular	36	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Gradient-boosted trees	lagged tabular	216	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Large MLP	legacy tabular	108	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
GRU variant	sequence	81	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config

Table A3: Standardized tuning protocol for the stronger-baseline comparison. All tunable families share the same split, validation criterion, and two-stage selection structure; the two simple heuristics remain fixed ex ante.

thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the full six-row Brier table plus one alternate-threshold and one log-loss robustness check.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.75	0.002	5/20
0.05	Reactive sharp	Reactive sharp	0.70	0.002	6/20
0.10	Reactive sharp	Calibrated baseline	0.55	0.002	9/20
0.25	Reactive sharp	Calibrated baseline	0.45	0.004	11/20
0.50	Reactive sharp	Calibrated baseline	0.30	0.013	14/20
1.00	Reactive sharp	Lagged smoother	0.05	0.061	19/20

Table A6: Forecast-side vs. deployed-side model selection in the full six-row event-micro benchmark summary. Under the fixed thresholding interface, deployed-suboptimal counts rise from 5/20 seeds at zero friction to 19/20 at friction 1.00.

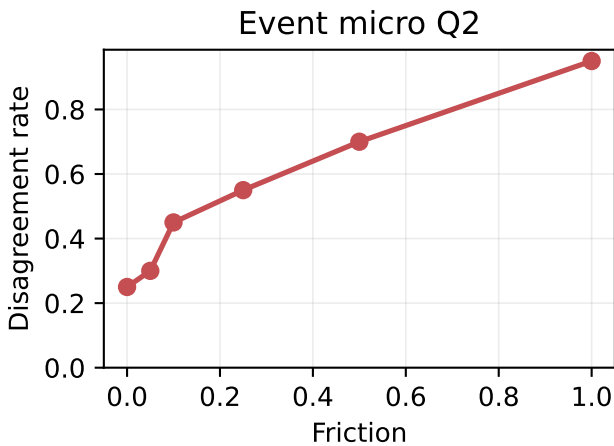


Figure A1: Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
$\tau=0.55$	0.00	Reactive sharp	Reactive sharp	0.75	0.002	0.000	5/20
$\tau=0.55$	0.50	Reactive sharp	Calibrated baseline	0.30	0.013	0.008	14/20
$\tau=0.55$	1.00	Reactive sharp	Lagged smoother	0.05	0.061	0.068	19/20
$\tau=0.50$	0.00	Reactive sharp	Reactive sharp	0.60	0.002	0.000	8/20
$\tau=0.50$	0.50	Reactive sharp	Lagged smoother	0.20	0.017	0.017	16/20
$\tau=0.50$	1.00	Reactive sharp	Lagged smoother	0.05	0.067	0.072	19/20

Table A7: Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.75	0.002	0.000	5/20
0.50	Reactive sharp	Calibrated baseline	0.25	0.013	0.010	15/20
1.00	Reactive sharp	Lagged smoother	0.05	0.061	0.068	19/20

Table A8: Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Across both appendix robustness checks, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in this deliberately small event-micro benchmark.

E Load-Following Support Domain

Using UCI ElectricityLoadDiagrams20112014, we build an aggregate multi-client load-following domain by summing disjoint client groups and evaluating at the selected 60-minute operational resolution. Groups 0 and 1 are used only for restricted calibration, while groups 2–9 serve as held-out evaluation units, so this appendix replaces the earlier household proxy with a stronger cross-sectional operational design.

A grouping-only balance-repair rerun substantially reduces group imbalance and Q1 target clipping while preserving the Q2 drift structure. A final restricted cap/margin search, however, finds no jointly feasible configuration that also sat-

Family	Representative	Validation mean	Validation median
Naive persistence	fixed ex ante	-2.911	-2.981
Moving average (7)	fixed ex ante	-3.265	-3.251
Linear AR	alpha=0.0001	-2.449	-2.512
Small MLP	lr=0.001, wd=0.0, batch=32	-2.331	-2.390
Small GRU	lr=0.001, batch=32	-2.628	-2.571
Regularized linear + lag search	lag=7, lasso, alpha=0.1	-2.418	-2.410
Gradient-boosted trees	lag=14, trees=100, depth=5, lr=0.1, leaf=10	-2.425	-2.279
Large MLP	w=256, d=3, drop=0.0, lr=0.001, wd=0.0001	-1.954	-2.012
GRU variant	L=28, h=128, layers=2, drop=0.0, lr=0.001	-2.602	-2.599

Table A4: Validation-selected representatives for the broader inventory Q2 family sweep. Family representatives are chosen by validation negative MAE only.

isfies the remaining Q1 requirement, so the domain remains appendix support.

Domain	Role	Groups	Res.	Q1	Q2	Balance	Note
Aggregate electricity-load proxy	support	8	60min	mixed support	promotion pass	balance_warn	appendix support only

Table A9: Load-following support-domain summary.

Under a fixed responsive interface, Q2 remains strong in this domain after balance repair, but Q1 high-friction behavior stays mixed and no jointly feasible configuration satisfies the remaining Q1 requirement.